

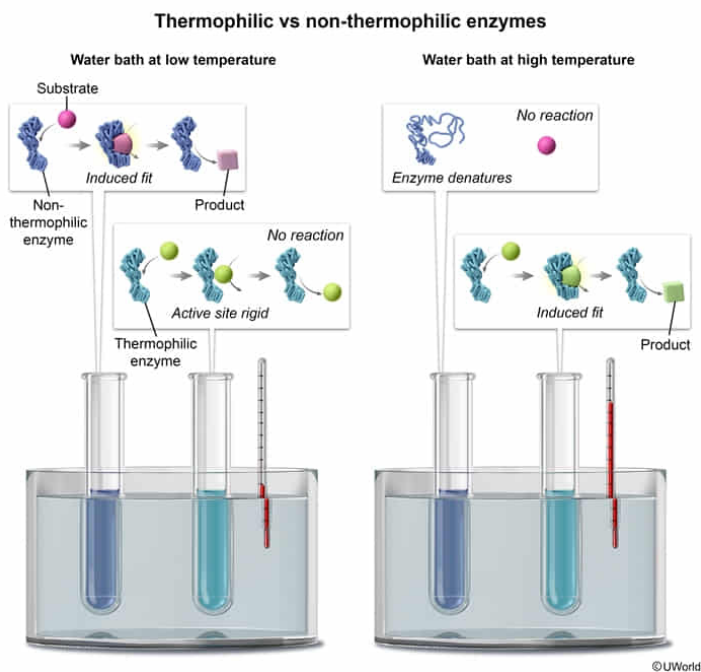
Thermophiles are microorganisms that thrive at temperatures above 40 °C. Which of the following characteristics LEAST accurately describes an enzyme isolated from a thermophile?

- ☐ A. Its optimal catalytic activity occurs above 40 °C. [4%]
- ☐ B. It is resistant to thermal denaturation. [22%]
- ☐ C. It has a relatively low catalytic rate at room temperature. [9%]
- ✓ ☒ D. It loses tertiary structure at temperatures below 37 °C. [63%]

Correct

63% answered correctly

## Explanation:



Enzymes are macromolecules (usually proteins) that catalyze reactions by stabilizing the transition state—a high-energy molecule midway between substrate and product—within a pocket of the enzyme known as the active site. This stabilization requires the enzyme's active site to change shape upon substrate binding (seen in the [induced fit model](#)) to accommodate the shape of the transition state. The ability of the active site to adopt and maintain the correct shape is **influenced by temperature**.

- At **low temperatures**, molecular motion slows down and the **active**

**site becomes inflexible.** Under these conditions the active site cannot easily change shape to accommodate the transition state, so the reaction rate is slow.

- At **high temperatures**, molecular motion speeds up and the intramolecular bonds that maintain the enzyme's **tertiary structure** are broken. This causes **proteins to denature**, destroying the shape of the enzyme's active site and halting catalytic activity.

To facilitate survival, **enzymes have evolved to operate** at the environmental **temperatures to which organisms are exposed**.

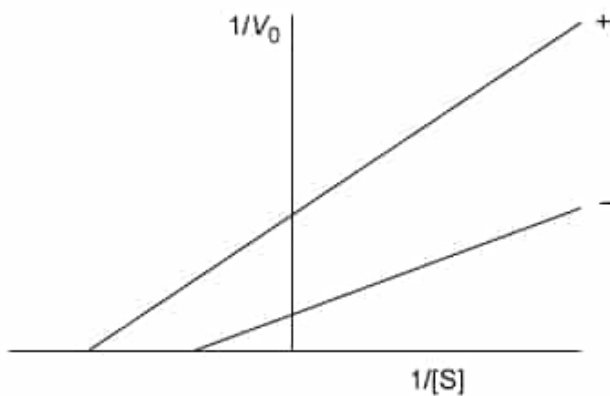
According to the question, thermophiles are microorganisms that thrive at temperatures above 40 °C. Therefore, enzymes from thermophiles typically have **optimal activity** above 40 °C and most likely have a relatively low catalytic rate at room temperature (~25 °C) (**Choices A and C**). Thermophilic enzymes must be resistant to thermal denaturation so that they can maintain their tertiary structures at the high temperatures of their natural environment (**Choice B**). Note that the term "resistant" does not mean these enzymes can never be thermally denatured, but rather that they *resist* thermal denaturation at temperatures that would denature proteins from non-thermophilic organisms.

The question asks for the LEAST accurate description. Although enzymes from thermophiles typically operate slowly below 37 °C, this is most likely due to inflexibility of the active site, not denaturation. Accordingly, **thermophilic enzymes most likely maintain their tertiary structure below 37 °C** because molecular motion at this temperature is too slow to break intramolecular bonds.

### **Educational objective:**

Enzymes operate optimally at a temperature that is high enough for the enzyme to maintain flexibility but low enough that the enzyme does not denature. Most enzymes have evolved to operate optimally at the temperatures to which they are typically exposed in their host organisms.

An enzyme was assayed in the absence (–) or presence (+) of an allosteric effector, and the following Lineweaver-Burk plot was obtained:



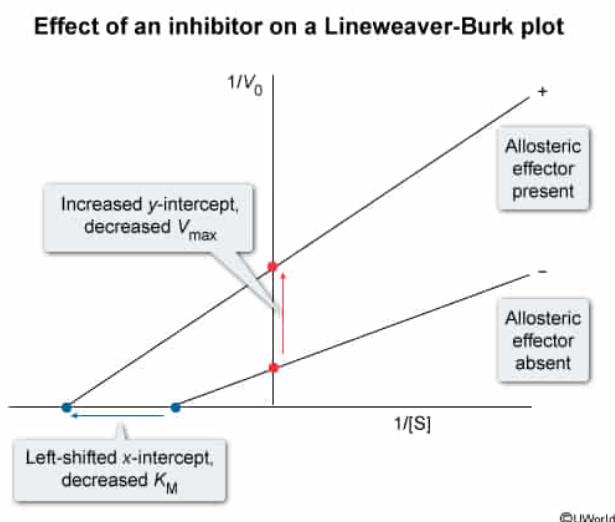
Which statement about the allosteric effector is correct?

- ☐ A. The effector is an activator that increases both  $V_{\max}$  and  $K_m$ . [16%]
- ☐ B. The effector is an activator that increases  $V_{\max}$  but decreases  $K_m$ . [16%]
- ☒ C. The effector is an inhibitor that decreases both  $K_m$  and  $V_{\max}$ . [49%]
- ☐ D. The effector is an inhibitor that increases  $K_m$  but decreases  $V_{\max}$ . [17%]

Correct

50% answered correctly

### Explanation:



Lineweaver-Burk plots display **enzyme kinetic data** in **linear form**, with the inverse of reaction velocity ( $1/V_0$ ) on the y-axis and the inverse of substrate

concentration ( $1/[S]$ ) on the x-axis. Establishing the location of the x- and y-intercepts on these graphs facilitates the determination of the **Michaelis-Menten parameters**  $V_{\max}$  (the maximum possible reaction rate) and  $K_m$  (the concentration of substrate at which  $\frac{1}{2}V_{\max}$  is achieved). The **y-intercept** of a Lineweaver-Burk plot is  $1/V_{\max}$ , and the **x-intercept** is  $-1/K_m$ .

**Allosteric effectors** are molecules that bind to enzymes at a site other than the active site and either increase activity (activators) or decrease activity (inhibitors). Lineweaver-Burk plots generated in the presence and absence of an allosteric effector can be compared to determine how the effector alters  $K_m$  and  $V_{\max}$ . Because the x- and y-intercepts are related to the **inverse of  $K_m$  and  $V_{\max}$** , an *increase* in the magnitudes of the intercepts (ie, an *increase* in their **absolute values**, or distances from the origin) corresponds to a *decrease* in  $V_{\max}$  and  $K_m$ , and vice versa.

The graph shows that in the presence (+) of the allosteric effector, the y-intercept increases, indicating a decreased  $V_{\max}$  and making the effector an inhibitor. Different inhibitors affect  $K_m$  in **different ways**. In this case, the x-intercept shifts leftward. Because the x-intercept is negative, this shift moves the intercept *away* from the origin, meaning its magnitude *increases*. The inverse relationship between the x-intercept and  $K_m$  therefore indicates that  $K_m$  has *decreased*. Taken together, the allosteric effector is an **inhibitor** that **decreases both  $V_{\max}$  and  $K_m$** .

**(Choices A and B)** On a Lineweaver-Burk plot, an **activator** would increase  $V_{\max}$  and cause a *decreased* y-intercept (shifting downward, toward the origin).

**(Choice D)** Inhibitors that **increase  $K_m$**  would appear as a *rightward shift* in the x-intercept on a Lineweaver-Burk plot (ie, decreased magnitude, shifting toward the origin).

### **Educational objective:**

Lineweaver-Burk plots are linear graphs that facilitate determination of the Michaelis-Menten parameters  $V_{\max}$  and  $K_m$ . The y-intercept is equal to  $1/V_{\max}$ , and the x-intercept is equal to  $-1/K_m$ . Upward shifts in the y-intercept correspond to a decreased  $V_{\max}$ , and leftward shifts in the x-intercept correspond to a decreased  $K_m$ .

**Time spent:**0 sec

**Subject:**  
Biochemistry

**QID:**402507

**System:**  
Enzymes